

Jay Bharambe

Data Scientist • ML / AI Engineer

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[Portfolio](#) | [GitHub](#) | [LinkedIn](#)

SUMMARY

Data Scientist with hands-on experience in programmatic AdTech, applying machine learning, deep learning, NLP, and causal inference to large-scale data with PySpark. Built production analytics that unlocked new client revenue and measurable performance gains, working across the stack from PyTorch model development and LLM fine-tuning to statistical modelling. Seeking a Data Scientist, ML Engineer, or AI Engineer role building impactful, scalable AI products.

SKILLS

Languages	Python, SQL, Dart
ML / Deep Learning	PyTorch, scikit-learn, HuggingFace Transformers, CNNs, model fine-tuning
NLP / LLMs	LangChain, LLM fine-tuning, prompt engineering, mBART, XLM-R
Stats / Causal Inference	Regression, classification, mixed-effects modelling, Difference-in-Differences (DiD)
Data at Scale	PySpark, Databricks, FAISS, Pandas, NumPy
Viz	Matplotlib, Seaborn
Backend / DevOps	Flask, Git, GitHub/GitLab, Bash, SSH/SCP, Jupyter, VS Code

WORK EXPERIENCE

Data Science Intern — MiQ Digital, Bengaluru Jan 2026 – Jun 2026

- Built the **reach-curve estimator** for the UK region; adopted by clients and **unlocked ~\$100K in new revenue**.
- Designed a high-performance **FAISS** similarity-search pipeline to cluster and match UK postcodes across multi-dimensional economic and population data, **raising match accuracy from 85% to 95%**.
- Delivered **incremental KPI lift measurement via Difference-in-Differences (DiD)** across all UK-region clients, providing causal uplift evidence that improved client retention and stickiness.
- Performed **mixed-effects modelling** to estimate client-level effects across hierarchical campaign data.

Android Development Intern — SimDaaS, Kanpur May 2025 – Dec 2025

- Built a Flutter app to remotely control a real-time autonomous IoT sprayer over MQTT, plus internal procurement and leave-management tools backed by Supabase (SQL) and Firebase (NoSQL).

PROJECTS & RESEARCH

CAMBART: Character-Aware mBART for Hinglish NLP — BS Thesis — IISER Bhopal Jan 2026 – Apr 2026

- Developed **CAMBART**, a character-aware multilingual NLP framework for code-mixed Hinglish, built on frozen mBART-50 and XLM-R backbones with lightweight character-injection adapters (PyTorch).
- Designed and evaluated three fusion architectures (scalar-gated, cross-attention, multi-scale gated) over a shared CharCNN encoder across translation, semantic parsing, summarization, and classification.
- Built a multi-task evaluation and noise-robustness pipeline with synthetic character perturbations, multi-seed experimentation, and visualization-driven analysis for low-resource multilingual NLP. ([GitHub](#))

Research on LLM Capabilities for Complex Semantic Tasks — ISL Lab, IISER Bhopal May 2025 – Jul 2025

- Benchmarked multiple LLMs on humor and code-mixed text detection, designing prompt-engineering pipelines that improved semantic transfer across English–Hindi, with applications in content moderation and cross-lingual translation.
- Evaluated LLM machine-translation quality on code-mixed input, identifying systematic failure modes; also applied LLMs to hate-speech detection in code-mixed sentences. ([GitHub](#))

Pix2Struct Fine-Tuning — Image-to-HTML Generation

- Fine-tuned the Pix2Struct vision-language model on the WebSight dataset to generate HTML directly from webpage screenshots, demonstrating multimodal sequence-to-sequence modelling. ([GitHub](#))

Graph-Based Movie Recommendation System

- Built a recommendation engine modelling users and movies as graph nodes; combined collaborative and content-based filtering and applied the Louvain algorithm for community detection. ([GitHub](#))

EDUCATION

B.S. in Data Science, Indian Institute of Science Education and Research (IISER), Bhopal 2022 – 2026

Higher Secondary (+2), Jawahar Navodaya Vidyalaya, Shegaon 2021

ACHIEVEMENTS & ACTIVITIES

- Finalist, Smart India Hackathon 2023 • Rank 3, Status Code 2 (Vultr Track), 2025 • Finalist, Armacode 0, 2025.
- Senior Developer, Coding & Networking Club, IISER Bhopal (2024–25) • Developer, VISTAR Cell, Dept. of Data Science & Engineering, IISER Bhopal (2025–26).